
Software Requirements Specification

for

Library Management System

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

A Library Management System (LMS) is a software application that simplifies and automates the operations of libraries. It is a complete system for managing library duties such as purchases, member management, monitoring, storing, and circulation. The primary objective of an LMS is to properly organize and manage the resources available in a library, making it easier for librarians to conduct everyday operations and create a user-friendly experience for users.

1.1 Purpose

This is the Software Requirements Specification (SRS) for the Library Management System. The purpose of this document is to convey information about the application's requirements, both functional and non-functional, to the reader. This document provides (a) a description of the environment in which the application is expected to operate, (b) a definition of the application's capabilities, and (c) a specification of the application's functional and nonfunctional requirements.

1.2 Document Conventions

The following conventions are used throughout this document:

- **Bold text** indicates section headings and key concepts.
- *Italicized text* is used for emphasis or reference to other documents.
- Uses Times New Roman font, with 16 pt bold for main headings, 14 pt bold for subheadings, and 12 pt regular for body text

1.3 Intended Audience and Reading Suggestions

This document is intended for the following readers:

- **Users** – who can also borrow books and may have extended privileges.
- **Librarians/Admins** – who manage book inventory, issue/return books, and handle library operations and oversee the entire system, including managing user roles and access control.

1.4 Project Scope

- ✓ The system is a digital solution that automates key library functions such as book, user, and transaction management, eliminating manual record-keeping.
- ✓ It supports three main user roles: Admin (Librarian), Users, with the Admin handling both user management and library operations.

✓ Admin (Librarian) can:

- Add, update, or delete books from the catalog.
- Issue and return books to/from users while tracking due dates and fines.
- Register and manage user accounts.
- Generate reports on issued books, overdue books, and system usage.
- Search, filter, and view the complete book inventory.

✓ Users can:

- Search and browse books by title, author, or category.
- Request to borrow or return books.
- View personal borrowing history, due dates, and any applicable fines.

1.5 References

The following documents and standards were used or referred to while creating this SRS.

1. **IEEE 830-1998** – IEEE Recommended Practice for Software Requirements Specifications
2. **Institution Library Policy Manual**, Rev 2.1, XYZ University, 2024
3. *Garcia-Molina, Ullman, Widom – Database Systems: The Complete Book*, 2nd Edition
4. *Ben Shneiderman – Designing the User Interface*, 6th Edition
5. MySQL 8.0 Documentation
6. FastAPI Documentation – Accessed June 2025

2. Overall Description

2.1 Product Perspective

The Library Management System (LMS) is a new, self-contained software product designed to automate and streamline library operations, replacing traditional manual record-keeping. It centralizes library resources, user management, and transaction processing, providing a digital interface for both library staff and users.

While LMS is a standalone system, it can be integrated with institutional databases for student/teacher records or with external reporting tools for analytics. It provides role-based access for admins and users, ensuring secure and efficient operations.

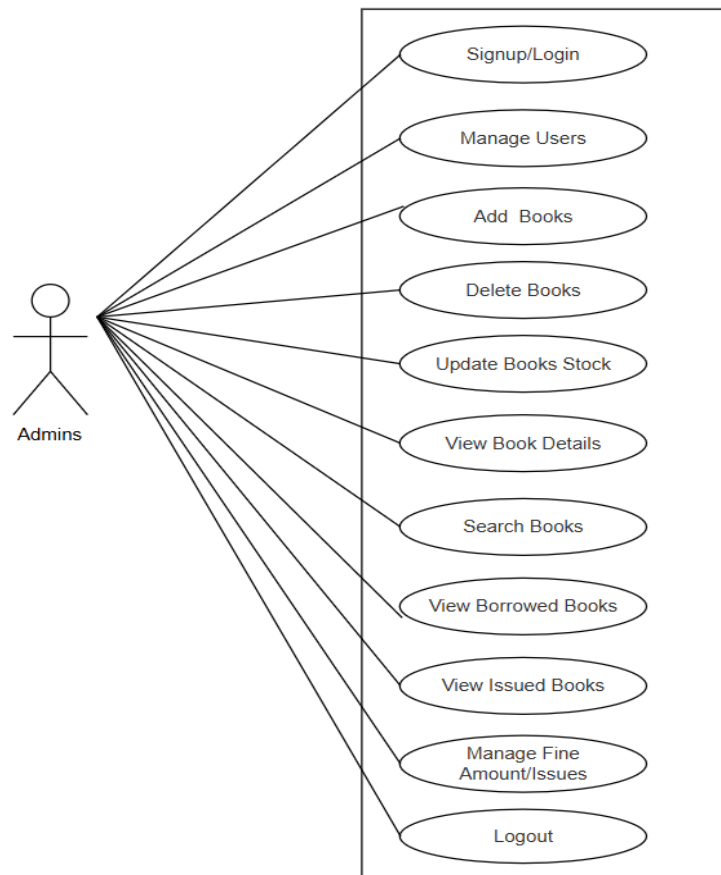
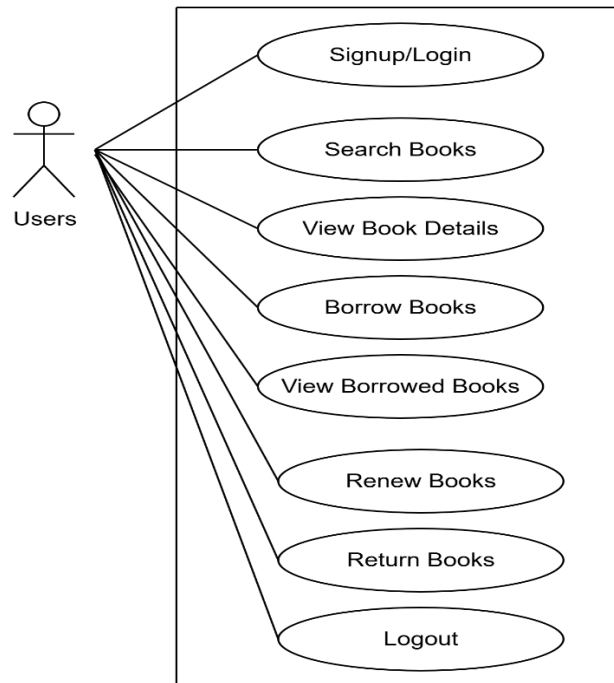
2.2 Product Features

The LMS provides the following major features:

- **User Management:**
 - Create, update, and delete user accounts (Admin only).
 - Role-based access: Admin and Users .
- **Book Management:**
 - Add, update, or delete books in the catalog.
 - Search and filter books by title, author, or category.
- **Circulation Management:**
 - Borrow and return books.
 - Track due dates, overdue books, and fines.
- **Reports & Analytics:**
 - Generate reports on issued books, overdue books, and system usage.
- **Dashboard Views:**
 - Users: View borrowing history, current loans, and due dates.
 - Admin: Monitor all library operations and user activity.

2.3 User Classes and Characteristics

USE CASE DIAGRAM



2.3.1 USE CASE DESCRIPTION

Use Case 1: Borrow Book

Actor: Student / Teacher

Precondition: User is logged in and eligible to borrow books

Trigger: User selects a book and chooses to borrow it

Main Success Scenario:

1. User searches and finds a book.
2. Views book details and availability.
3. Selects the “Borrow” option.
4. System checks borrowing eligibility and book availability.
5. System updates status and adds it to the user's borrowed list.
6. Due date is generated and displayed.

Exception Scenarios:

- **Book not available:** → System shows “Not Available”
- **Max borrow limit reached:** → System restricts further borrowing
- **User has overdue books:** → Borrowing blocked until returns/fines are cleared

Use Case 2: Return Book

Actor: Users

Precondition: Book is already borrowed by the user

Trigger: User chooses to return a borrowed book

Main Success Scenario:

1. User opens their borrowed books list.
2. Selects a book to return.
3. System records return and updates book status.
4. Return confirmation and updated borrowing count shown.

Exception Scenarios:

- **Book already returned:** → System shows status as “Returned”
- **System error:** → Logs issue and notifies admin

Use Case 3: Renew Book

Actor: Student / Teacher

Precondition: Book is currently borrowed and within renewal period

Trigger: User chooses to renew a borrowed book

Main Success Scenario:

1. User opens borrowed books section.
2. Selects “Renew” for an eligible book.
3. System checks due date and allows renewal.
4. Due date is extended and updated.

Exception Scenarios:

- **Book is on hold by another user:** → Renewal blocked
- **Book renewal limit reached:** → System denies renewal

Use Case 4: Add Book Resource

Actor: Admin

Precondition: Admin is logged in

Trigger: Admin wants to add a new book to the library

Main Success Scenario:

1. Admin selects “Add Book” option.
2. Enters book details (title, author, ISBN, quantity).
3. System validates and saves data.
4. Book becomes searchable and borrowable.

Exception Scenarios:

- **Duplicate entry detected:** → System shows warning
- **Missing fields:** → System prompts for completion

Use Case 5: Manage Fines & Issues

Actor: Admin

Precondition: Admin is logged in

Trigger: Admin reviews overdue books or user complaints

Main Success Scenario:

1. Admin views overdue reports.
2. Applies fines or adjusts due dates.
3. Marks books as lost/damaged if needed.
4. Users are notified of penalties.

Exception Scenarios:

- **Invalid fine amount entered:** → System alerts and halts action
- **Dispute raised by user:** → Forwarded for admin resolution

Use Case 6: Search and View Book Details

Actor: Student / Teacher / Admin

Precondition: User is logged in

Trigger: User performs a book search

Main Success Scenario:

1. User enters keywords or filters.
2. System fetches matching results.
3. User selects a book to view full details.

Exception Scenarios:

- **No books found:** → System shows “No results”

Use Case 7: Manage Users

Actor: Admin

Precondition: Admin is logged in

Trigger: Admin wants to add, edit, or remove users

Main Success Scenario:

1. Admin opens user management portal.
2. Adds or edits user roles (Users/Admin).
3. System updates user data and access rights.

Exception Scenarios:

- **User not found:** → System shows “User not available”
- **Unauthorized change attempt:** → Logs activity and blocks action

2.4 Operating Environment

- **Hardware:**
 - Minimum: Dual-core processor, 4 GB RAM, 20 GB storage.
 - Recommended: Quad-core processor, 8 GB RAM, 50 GB storage.
- **Software:**
 - Backend: FastAPI / Python
 - Database: MySQL
 - Frontend: HTML, CSS, JavaScript
 - OS: Windows 11
 - Browser: Firefox

2.5 Design and Implementation Constraints

- Must use MySQL for database operations.
- Backend implemented in FastAPI/Python.
- Frontend limited to HTML, CSS, and JavaScript.
- Enforce role-based access control.
- Comply with institutional data privacy policies.
- System must support up to 1000 concurrent users without performance issues.

2.6 User Documentation

- User Manual: Instructions for general users to browse, borrow, and return books.
- Admin Guide: Instructions for Admin to manage books, users, and generate reports.

2.7 Assumptions and Dependencies

Assumptions:

- Users have basic computer literacy.
- Stable internet connection is available.
- Each user has a unique account for authentication.

Dependencies:

- Availability of MySQL server for database operations.
- Access via a web browser.
- Compliance with institutional policies regarding borrowing, fines, and user management.

3.1 User Authentication Module

3.1.1 Description and Priority

- Provides secure login and registration for Admins and Users.
- Ensures role-based access control.
- Priority: High (critical for system access and security).

3.1.2 Stimulus/Response Sequences

- User enters credentials → System validates and grants access.
- Admin/User registers with details → System stores profile in the database.
- Forgotten password request → System sends reset instructions.

3.1.3 Functional Requirements

- System shall allow users (admin/user) to log in using a valid username and password.
- System shall enforce role-based permissions (admins vs. users).
- System shall provide a registration option for new users.
- System shall support password recovery using email/username verification.

3.2 Book Management Module (Admin)

3.2.1 Description and Priority

- Admins can manage the book catalog by adding, updating, or removing records.
- Priority: High (essential for keeping the catalog up to date).

3.2.2 Stimulus/Response Sequences

- Admin enters new book details → System saves and updates catalog.
- Admin deletes a book → System removes it and adjusts stock.
- Admin updates book info → System reflects changes in database.

3.2.3 Functional Requirements

- System shall allow admins to add books with title, author, and stock quantity.
- System shall allow admins to update existing book details.
- System shall allow admins to delete outdated/unavailable books.
- System shall allow searching books by title (admin view).

3.3 Book Search Module (Users and Admins)

3.3.1 Description and Priority

- Provides search and filter features for both users and admins.
- Priority: High (core user-facing functionality).

3.3.2 Stimulus/Response Sequences

- User enters book title → System displays results with availability.
- Admin searches book → System displays details for management actions.

3.3.3 Functional Requirements

- System shall allow search by book title.
- System shall display book details (title, author, availability).
- System shall indicate if the book is available for borrowing.

3.4 Borrowing and Returning Module

3.4.1 Description and Priority

- Manages book circulation: issuing, returning, and updating stock.
- Priority: High (main service offered by LMS).

3.4.2 Stimulus/Response Sequences

- User requests to borrow a book → System checks stock → If available, issue record created.
- User returns book → System updates stock and checks overdue fines.

3.4.3 Functional Requirements

- System shall allow users to borrow books if stock is available.
- System shall track borrowing details (issue date, due date, user ID, book ID).
- System shall update stock upon borrowing or returning.
- System shall allow admins to verify book returns.

3.5 Fine Management Module (Admin)

3.5.1 Description and Priority

- Calculates and informs users about overdue fines.
- Priority: Medium (important but secondary to borrowing/returning).

3.5.2 Stimulus/Response Sequences

- User returns book late → System calculates fine → System notifies user during return process.

3.5.3 Functional Requirements

- System shall calculate fines for overdue returns based on predefined rules.
- System shall notify users of fine amount during return.
- System shall allow admins to update or waive fines if needed.

4. External Interface Requirements

4.1 User Interfaces

- **Frontend (Client-Side):**
 - Languages: HTML5, CSS3, JavaScript (Vanilla JS)
 - Browser: Google Chrome (latest version recommended)
 - Features: Responsive interface for desktop and mobile; intuitive forms for search, borrow, return, and reports.

4.2 Hardware Interfaces

- **Development Machine:**
 - Processor: AMD Ryzen 7 5800H
 - RAM: 16 GB
 - Storage: 512 GB SSD or more
 - OS: Windows 11 Home 64-bit
- **Server Requirements :**
 - Processor: Quad-core 2.5 GHz or higher
 - RAM: 8 GB or higher
 - Storage: 256 GB SSD or higher
 - Network: Stable internet connection for API services

4.3 Software Interfaces

- **Backend (Server-Side):**
 - Programming Language: Python 3.10+
 - Framework: FastAPI
 - Web Server: Uvicorn (for development)
 - Database: MySQL
- **Frontend (Client-Side):**
 - Languages: HTML5, CSS3, JavaScript
 - Browser Compatibility: Chrome, Edge, Firefox (latest versions recommended)

- **Development Tools:**
 - IDE/Editor: VS Code
 - Database GUI: MySQL Workbench

4.4 Communications Interfaces

- **Protocols & Standards:**
 - HTTP/HTTPS for client–server communication (browser ↔ FastAPI API)
 - SMTP API for sending email notifications (due date reminders, fines)
 - MySQL Protocol (TCP/IP, port 3306) for backend–database communication
- **Message Formatting:**
 - JSON for API request/response data
 - HTML/CSS/JS for frontend rendering
- **Security & Encryption:**
 - HTTPS (TLS 1.2/1.3) for secure data transfer
 - Sensitive data (passwords, user details) encrypted and hashed before storage
- **Data Transfer Requirements:**
 - Minimum 2 Mbps bandwidth for smooth API and database communication
 - Sync/retry mechanism to handle temporary network disconnections (borrow/return actions)

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time**
 - API requests (search, borrow, return): **< 2 seconds**.
 - Report generation (overdue, borrowed history): **< 8 seconds**.
- **Scalability**: System should support up to **2000 concurrent users** (students, teachers, admins).
- **Availability**: 99.5% uptime during library hours (**8 AM – 8 PM**).
- **Notifications**: Email/SMS reminders delivered within **5 seconds**.

5.2 Safety Requirements

- Prevent **accidental deletion** of book and user records (confirmation prompts, soft deletes).
- Restrict **file uploads** (e.g., book covers, reports) to safe formats: PNG, JPG, PDF.
- Input validation for **ISBN, dates, user IDs** to avoid invalid records.
- Backup system ensures **daily database backups** with recovery option.

5.3 Security Requirements

- **User authentication** (student, teacher, admin) with **role-based access control**.
- **Data encryption** during transmission (HTTPS) and hashed storage (bcrypt).
- **Password policy**: strong, stored as salted hashes.
- **Audit logging**: All user/admin actions logged in a tamper-proof manner.
- **Compliance**: Follow local data protection/privacy laws.

5.4 Software Quality Attributes

- **Usability:** Intuitive web interface requiring minimal training.
- **Reliability:** Stable handling of **2000 concurrent users** without crashes.
- **Maintainability:** Modular backend (FastAPI routes, models, services).
- **Portability:** Works on **major browsers (Chrome, Edge, Firefox)** and mobile-friendly (responsive design).
- **Interoperability:** REST APIs support integration with external systems (attendance, ERP).
- **Testability:** APIs testable using **Postman / Swagger docs**.
- **Robustness:** Handles invalid input gracefully with validation and error messages.

6. Other Requirements

6.1 Database Requirements

- The system shall use MySQL 8.0 with normalized structure (3NF) and enforced referential integrity.
- Daily automated backups must be maintained with recovery options to prevent data loss.

6.2 Internationalization Requirements

- The system shall support Unicode (UTF-8) for multilingual book titles, author names, and user details.
- Default language is English, with scope for future support of local languages (e.g., Hindi, Tamil).

6.3 Legal and Compliance Requirements

- The system must comply with local data protection and privacy laws, ensuring secure handling of user data.
- Only metadata and cover images of books may be stored, respecting copyright regulations.

6.4 Reuse Objectives

- The system shall adopt a modular architecture (frontend, backend, database) for easy component reuse.
- Developed APIs for book search, issue/return, and reporting shall be reusable by external systems.

6.5 Future Enhancements

- Support for RFID/barcode scanners for automated check-in/check-out.
- Development of a mobile application for enhanced user convenience.

Appendix A: Glossary

Term	Definition
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Admin	Librarian with full system privileges (manage books, users, transactions).
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User	A student or teacher who can search, borrow, and return books.
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ISBN	International Standard Book Number used for unique identification of books.
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Overdue	A borrowed book that has not been returned by the due date.
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Fine	Monetary penalty applied for overdue books.
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LMS	Library Management System.
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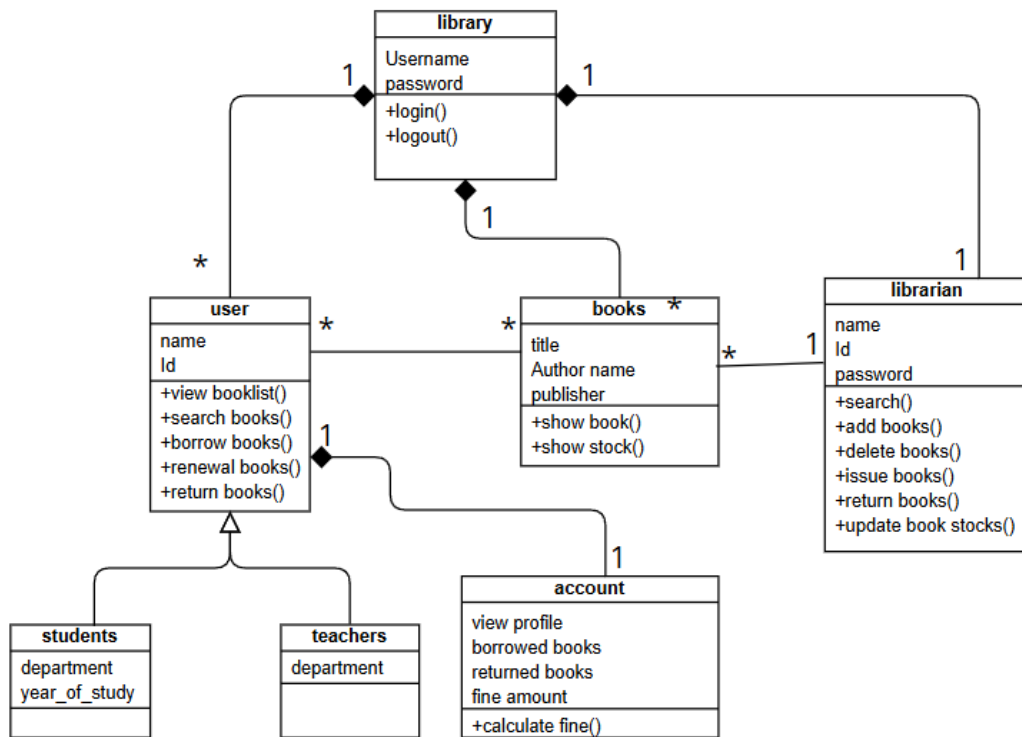
API	Application Programming Interface, enabling communication between frontend and backend.
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CRUD	Create, Read, Update, Delete – operations for managing books and users.
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Appendix B: Analysis Models

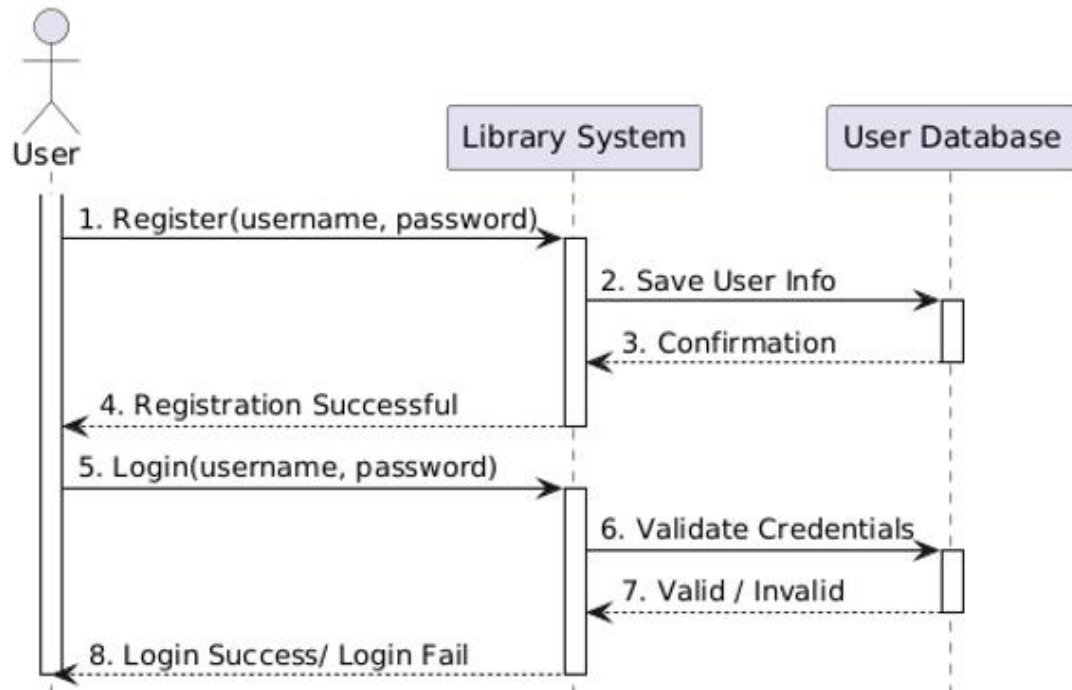
FUNCTIONAL REQUIREMENTS

CLASS DIAGRAM:

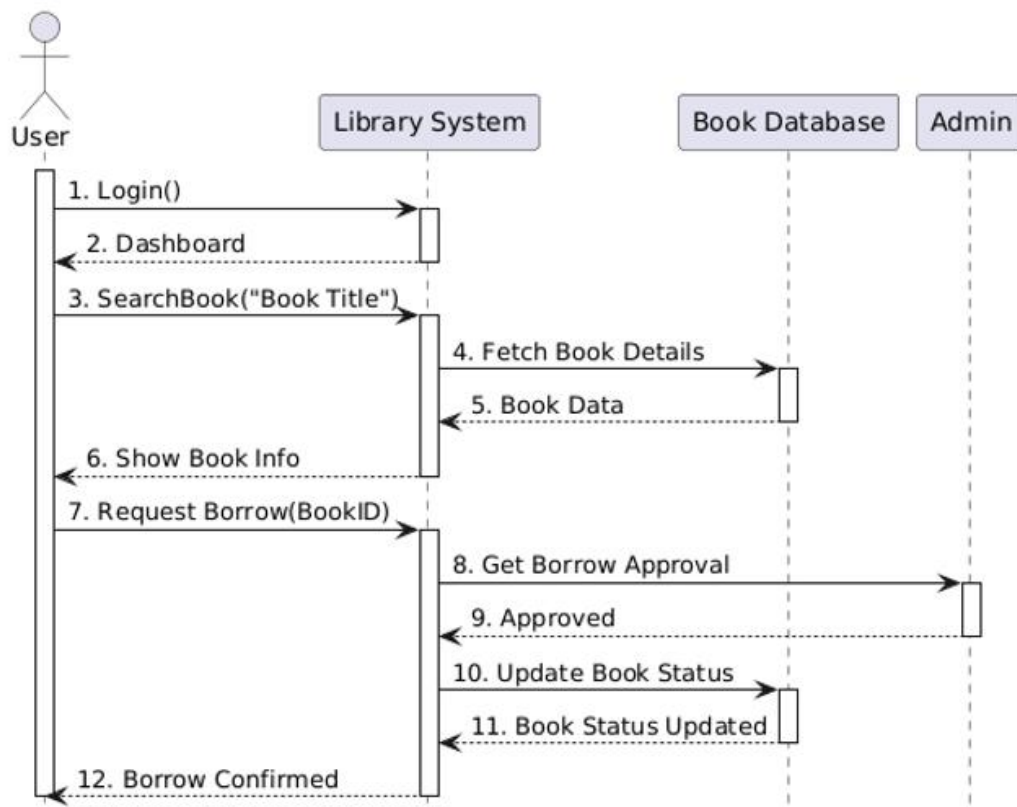


SEQUENCE DIAGRAM

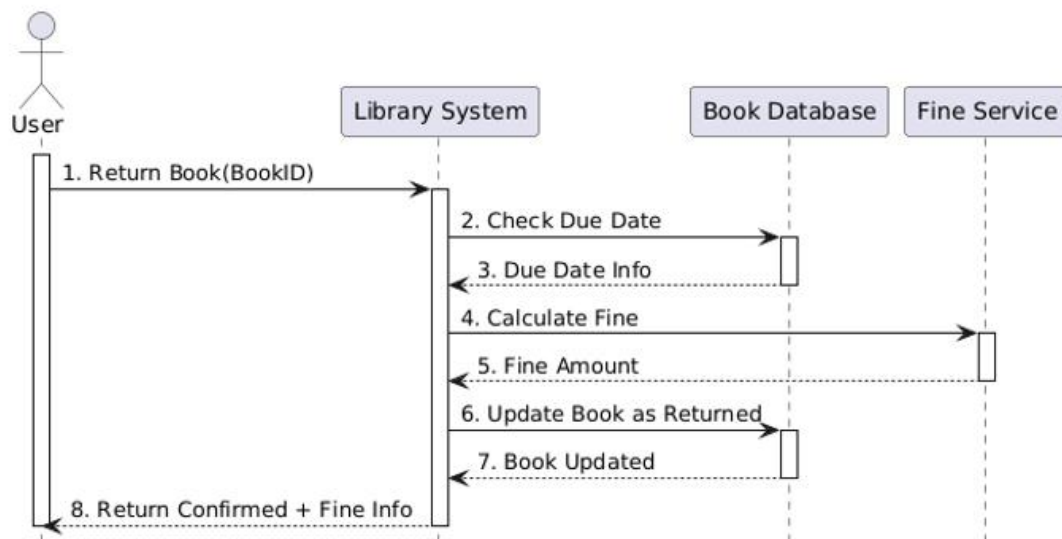
User authentication



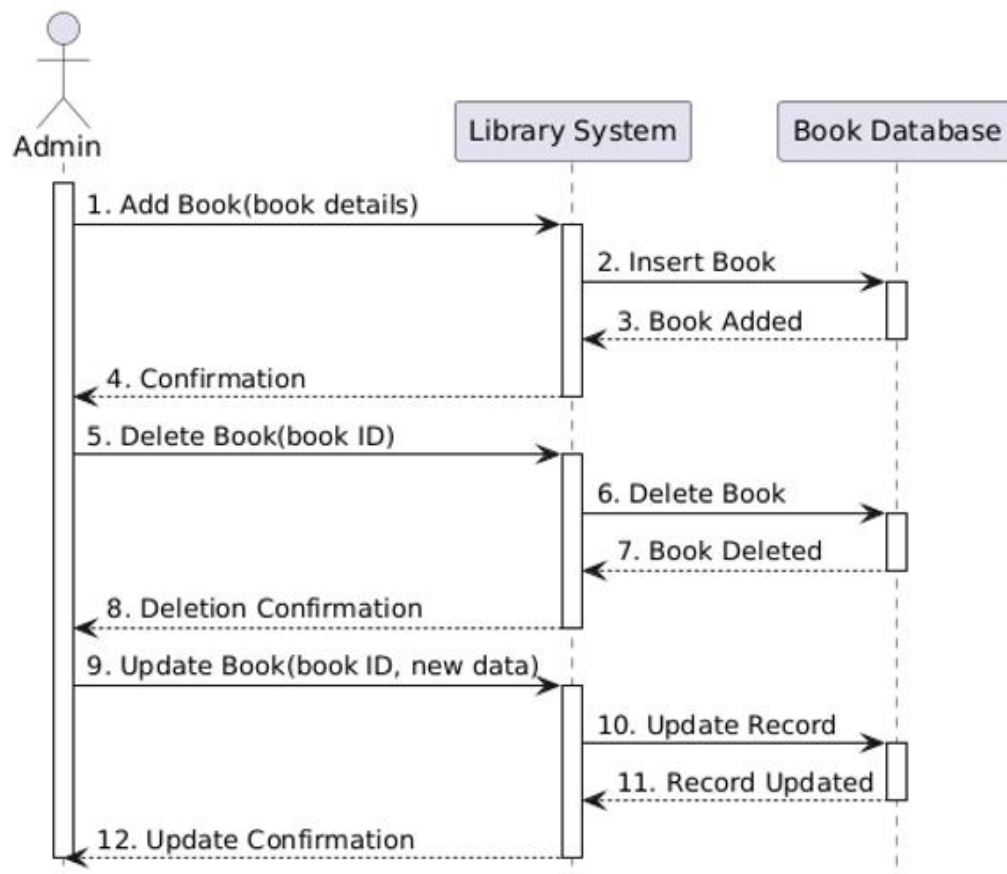
Request and Borrow Books Process



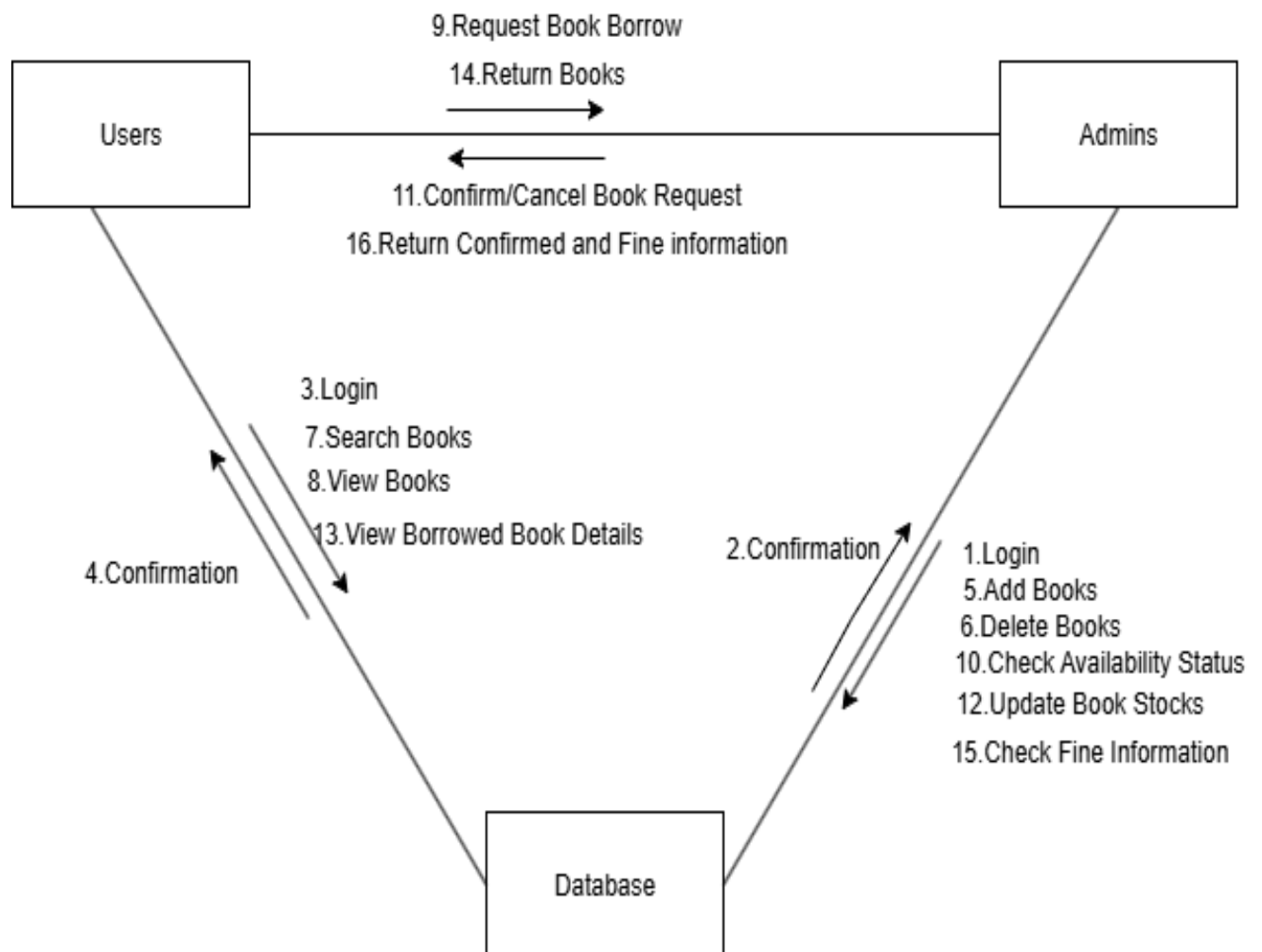
Return Books and Fine Service



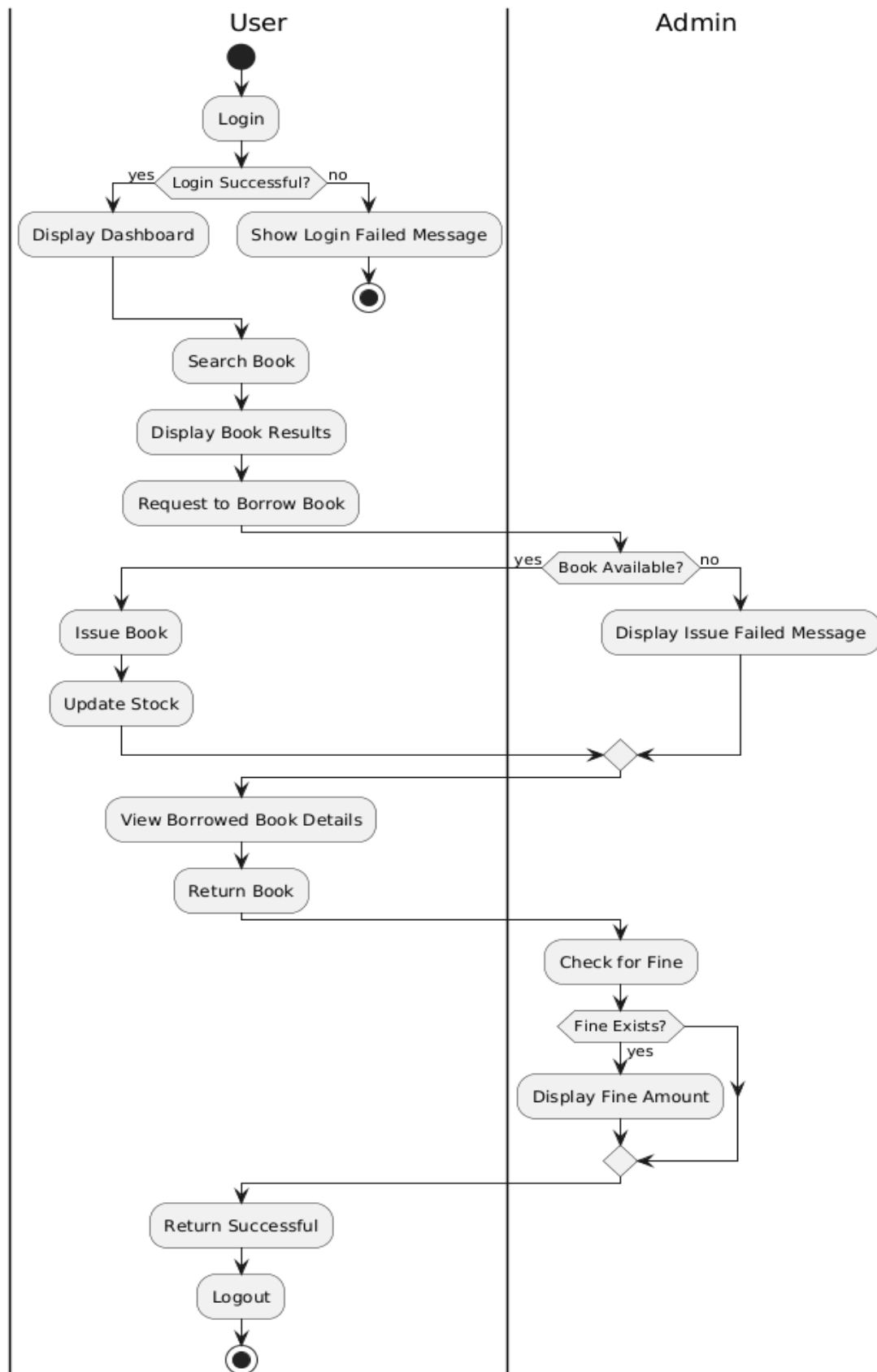
Add, Delete, Update Book Stocks



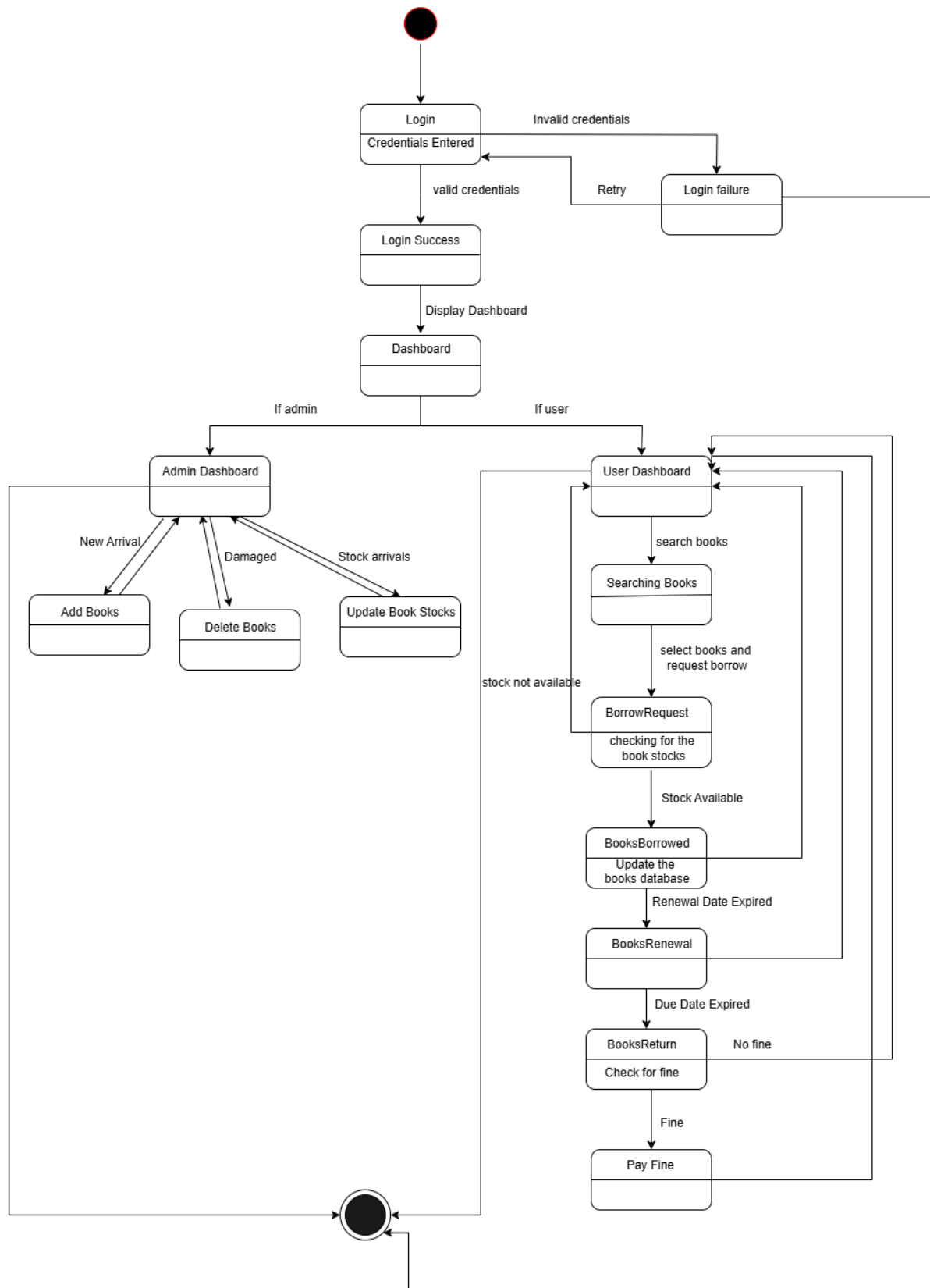
COLLABORATION DIAGRAM



ACTIVITY DIAGRAM



STATE CHART DIAGRAM



Appendix C: Issues List

- **Fine Calculation Policy** – Decision pending on whether fines will be calculated per day overdue or as a flat fee.
- **System Integration** – Scope of integration with external systems (ERP, attendance) is yet to be finalized.
- **Internationalization** – Languages other than English (e.g., Hindi, Tamil) need to be confirmed for future support.
- **RFID/Barcode Support** – Decision required on whether RFID/barcode scanning to be included.